



Dynamics of Titan's high-pressure ice layer

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ABSTRACT

The presence of ^{40}Ar in Titan's atmosphere and the replenishment of methane argue for the exchange between the interior and the atmosphere. These observations triggered the present study that aims to determine the conditions under which the high-pressure (HP) ice layer, likely present between the deep ocean and the silicate core, poses a barrier for the exchange of volatiles. We model heat and water transport through this convecting HP ice layer using a two-phase numerical model of solid ice-liquid water mixture. We observe that for a large range of heat fluxes from the silicate core and HP ice layer thicknesses, a few percent of liquid water forms at the interface with the silicates. Liquid water being less dense than the HP ice, it creates additional buoyancy, thus facilitating the transport of volatiles towards the ocean. Our results also show that convection is characterized by the presence of hot and the absence of cold plumes. We derive a scaling law that describes the dependence of a critical heat flux for the onset of melting at the silicates interface on the thickness of the HP ice layer and the ice viscosity. We also study the processes at the interface with the base of the ocean where a few tens of kilometers thick layer of temperate (partially molten) ice is present. We find a scaling law for its thickness that depends mainly on the ice viscosity and the density difference between the ice and water. Water from this partially molten, temperate layer flows into the ocean thus completing the connection with the silicate core. The water flux depends primarily on the amount of heat supplied from the silicates. Future evolution models that will use the scaling laws derived in this study will place bounds on the timing of these exchange processes. Using Cassini data and reasonable values of HP ice viscosity and silicate heat flux, we predict melting at the silicates/HP ice interface at present time.

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1. Introduction

The Cassini mission to the Saturnian system has revolutionized our ideas about the interior dynamics and geological evolution of icy moons by documenting the strong habitable potential of Enceladus and Titan. On Enceladus, cracks in the ice crust connect the ocean to the surface and geysers expel the deep ocean into space (Porco et al., 2014). On Titan, the exchange processes between the interior and the surface must have existed in order to explain the presence of ^{40}Ar and the replenishment of methane (Atreya et al., 2006).

The moment of inertia derived from the degree 2 gravity coefficients (Iess et al., 2010; Durante et al., 2019) suggests that Titan is differentiated into a rocky core made of hydrated silicates (Castillo-Rogez and Lunine, 2010; Néri et al., 2020) and a hydrosphere composed of a high-pressure (HP) ice layer, an ocean, and an ice I crust. The HP ice layer may be composed of different phases (ice VI, ice

V, ice III, cf. Fig. 1, left). The presence of an ocean has been inferred from three independent measurements: First, the Huygens probe detected an electric signal at 36 Hz whose variations during the descent are best explained by the presence of a conductive layer, which may be the salty ocean, about 65 km below the surface (Beghin et al., 2012). Second, the radio science experiment onboard the Cassini spacecraft was able to infer a large value of the degree 2 tidal Love number that suggests a decoupling between the ice crust and the interior (Iess et al., 2012). Third, the obliquity of 0.5 deg is best explained by the presence of a deep ocean, less than 100 km thick (Baland et al., 2014). Note that the interpretations of the different observations all agree on the presence of a deep ocean but not on its depth, which is equivalent to the thickness of the ice crust that in turn constrains the HP ice layer thickness (cf. Appendix A).

Three observations point to exchange processes between the deep interior and the atmosphere. First, the large amount of ^{40}Ar that was measured in the atmosphere (Niemann et al., 2010) indicates its release from the silicate core which represents the most likely reservoir of ^{40}K , its parent isotope. Second, the methane ob-

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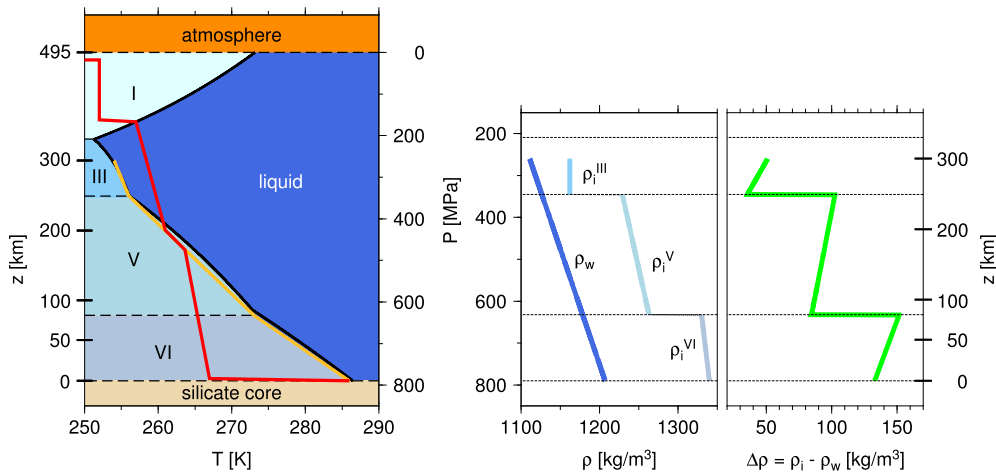


Fig. 1. Left: Titan's interior structure within the phase diagram of water ice (black lines, Wagner et al., 1994, 2011) - the pressure range indicates the necessity to take into account the phase change in the solid ice (the particular ice phases are marked with Roman numerals). The red curve shows a possible temperature profile through the hydrosphere corresponding to the reference HP ice layer thickness of 200 km and assuming adiabatic profile in the ocean and convection in the ice I crust. The yellow line shows the linear approximation of the melting temperature T_m used in this study (cf. Table 1). Middle: density profiles of ice (ρ_i) and water (ρ_w) used in the simulations. The horizontal dashed lines indicate the solid phase change. Right: Density difference between the phases, $\Delta\rho = \rho_i - \rho_w$. (For interpretation of the color(s) in the figure(s), the reader is referred to the web version of this article.)

served in Titan's atmosphere would be photolyzed in a few tens of millions of years (Yung et al., 1984), thus its present-day amount implies a replenishment process such as the destabilization of the methane clathrate rich crust by impacts and/or convection-related processes (Choukroun and Sotin, 2012; Tobie et al., 2006) at different times of Titan's evolution or a one time event that happened some 100s of millions years ago (Tobie et al., 2006). Methane clathrates can form by serpentinization of the silicates (Atreya et al., 2006). Third, based on the observation of $^{15}\text{N}/^{14}\text{N}$ isotope ratio, Miller et al. (2019) suggest that up to 50% of Titan's atmospheric N_2 comes from the decomposition of insoluble organic matter (IOM) present in Titan's silicate core (Néri et al., 2020). Here, we study the dynamics of Titan's HP ice layer treating it as a two-phase mixture to provide a scenario by which volatiles can be transferred from the silicate core through the HP ice layer to the ocean. We do not investigate the potential pathways to deliver the volatiles through the ocean and ice I crust on Titan's surface and into its atmosphere.

In a previous study applied to Ganymede's HP ice layer (Kalousová et al., 2018), we demonstrated that under certain conditions, water can be produced at the silicates/HP ice interface and transported efficiently into Ganymede's ocean. Because Titan is less differentiated than Ganymede, the thickness of its hydrosphere as well as its HP ice layer is smaller (Néri et al., 2020, and Appendix A). Consequently, the silicates/HP ice interface is located at smaller pressure, making other HP ice phases than ice VI important (Fig. 1, left). One aspect of this paper is to investigate the effect of phase transitions in the solid ice on the efficiency of convection in the HP ice layer and the presence of water at the silicates/HP ice interface. Titan also has a surprisingly large eccentricity (0.029) although there is no resonance to sustain it. Tobie et al. (2005) proposed that this eccentricity must be the remnant of a larger value whose origin is unknown and they also provided values of the associated tidal dissipation in the HP ice layer. Here, we include simulations with volumetric heating to investigate the effect of tidal dissipation on the convection pattern and other characteristics of the HP ice layer.

The structure of this paper is as follows: We briefly summarize the main features of our model in Section 2 with more details given in the Appendix. We describe the results of our simulations in Section 3 and discuss the implications for the transfer of

volatiles and the dynamics of the HP ice layer in Section 4. Our findings are summarized in Section 5.

2. Model

Titan's interior structure is determined following the approach described by Vance et al. (2014) for Ganymede, using the values of the total mass ($M = 1.3453 \times 10^{23}$ kg) and the moment of inertia ($C = 0.341$) of Titan (Durante et al., 2019). Another strong constraint comes from the presence of a global ocean inferred from several independent measurements as described above. We provide a detailed description of the method to construct Titan's interior structure model in the Appendix A. We obtain the value of $R_s = 2080$ km for the radius of the silicates/HP ice interface which is consistent with the values proposed by Castillo-Rogez and Lunine (2010) and Fortes (2012). It gives the total hydrosphere thickness of 495 km (cf. left panel of Fig. 1). Since the current HP ice layer thickness is highly uncertain, we investigate values between 50 and 300 km. Left panel of Fig. 1 shows the interior structure of Titan within the phase diagram of H_2O . For the HP ice layers thicker than 80 km, the phase change from ice VI to ice V has to be taken into account. For cooler hydrospheres with very thick HP ice layers ($H > 250$ km), another phase change from ice V to ice III occurs. The red curve shows the possible temperature profile through the hydrosphere corresponding to the HP ice layer thickness $H = 200$ km.

We treat the HP ice layer material as a mixture of two coexisting phases – solid ice and liquid water (Souček et al., 2014). In this approach, the amount of water is described by porosity ϕ that is defined as a volume fraction of water in the whole mixture (cf. Appendix B for governing equations). In our simulations, porosities are typically on the order of a few percent indicating that ice is the dominant phase in the mixture. Our model allows liquid water to be advected by the convecting solid ice as well as to percolate through the solid ice. We do not take into account any fracturing. In our recent studies tailored to the HP ice layer of Ganymede (Kalousová et al., 2018; Kalousová and Sotin, 2018), we assumed that the HP ice layer was only composed of ice VI. In the case of Titan, the phase change in the solid ice has to be taken into account (cf. Fig. 1). Some other physical processes were included here that were neglected in the previous study (cf. Appendix B for more details). We use finite element method implemented in the

Table 1List of parameters used in the present study. z gives the vertical coordinate in km. •* denotes the varied model parameters.

Symbol	Physical property	Value/range/equation				Unit
H^*	HP ice layer thickness	{50, 100, 200, 300}				km
q_s^*	heat flux from silicates	{5, 10, 20, 40}				mW/m ²
Q^*	internal heating	{0, 5e-11, 1e-10}				W/kg
g	gravity acceleration	1.49–0.00027 z				m/s ²
H₂O phase						
z	vertical coordinate	VI {0–80}	V {80–250}	III {250–300}	liquid {0–300}	km
ρ_i/ρ_w	ice/water density ^{a,b}	1340–0.125 z	1263–0.2 (z –80)	1162	1207–0.32 z	kg/m ³
T_m	depth-dependent melting temperature ^b	286–0.1625 z	273–0.1 (z –80)	256–0.04 (z –250)	-	K
L	latent heat of melting ^{b,c}	306–0.5125 z	275	256	-	kJ/kg
L_s	solid phase change latent heat ^{b,c}	1	1/4	4	-	kJ/kg
α_i/α_w	ice/water thermal expansion ^a	1.46e-4	1.46e-4	1.46e-4	2.07e-4	1/K
c_i^p/c_w^p	ice/water specific heat ^d	2650	2650	2650	3850	J/kg/K
k_i/k_w	ice/water thermal conductivity ^e	1.58	1.58	1.58	0.56	W/m/K
μ_0^i/μ_w	reference ice/water viscosity ^{f,g,h}	1e15	{1e15, 1e16, 1e17}	1e15	2e-3	Pa s
E	ice viscosity activation energy ^g	30	30	30	-	kJ/mol
K_0	reference permeability ^{i,j}	1e-9	1e-9	1e-9	-	m ²
n	permeability exponent ^{i,j}	2	2	2	-	-
ϕ_c	percolation threshold	1	1	1	-	%

^aBezacier et al. (2014), ^bBridgman (1912, 1937), ^cHobbs (1974), ^dTchijov (2004), ^eRoss et al. (1978), ^fDurham et al. (1996, 1997); Durham and Stern (2001), ^gSotin and Poirier (1987); Sotin et al. (1985), ^hLide (2004), ⁱGolden et al. (2007), ^jPetrich et al. (2006)

open source software FEniCS (Logg et al., 2012; Alnæs et al., 2015). All material parameters used in the present study are listed in Table 1 and described in more detail in Appendix C.

3. Results

We performed 15 two-dimensional simulations in a Cartesian geometry tailored to the conditions within the HP ice layer of Titan while varying several model parameters. In previous studies (Kalousová et al., 2018; Kalousová and Sotin, 2018), we already investigated the effect of the ice VI reference viscosity as well as the effect of the ice permeability law. In the present study, we focus on parameters that are important for Titan or those that were not studied before: HP ice layer thickness H , heat flux from the silicate core q_s , ice V reference viscosity μ_0^V , and internal heating Q that can be produced by tidal dissipation in the HP ice layer. Note that ice VI reference viscosity (viscosity at the melting temperature) is equal to 10^{15} Pa s in all simulations (cf. Table 1), but we use a range of 10^{13} – 10^{17} Pa s in the derivation of scaling laws.

Fig. 2 shows the steady-state results of the reference simulation with $H = 200$ km, $q_s = 20$ mW/m², and $\mu_0^V = 10^{15}$ Pa s. No internal heating was prescribed in this simulation ($Q = 0$ W/kg). Our results show that a continuous layer of temperate (partially molten) ice where temperature equals the melting temperature, $T = T_m$, is established at the top boundary (panel a). Melt is generated in the upwelling hot plumes as well as in the top temperate layer from where it is extracted into the ocean (panel b). Some melt may also be generated at the bottom boundary with the silicates. The amount of liquid water (porosity) everywhere in the HP ice layer is small and does not exceed a few percent. Panels c and d (average temperature profile) show that the HP ice layer of thickness H comprises three distinct domains – a layer of temperate ice (gray region) at the top interface with the ocean of thickness H_t (melting always occurs in this layer), a convective layer of thickness H_c (some melting may occur in the upwelling hot plumes), and the bottom boundary where another temperate layer of thickness H_b is present. Note that $H_b \ll H_t$. Depending on the presence or not of this temperate layer H_b , melting may or may not occur at the silicates interface. If melt appears, it provides an additional buoyancy since liquid water is less dense than the solid HP ice. This additional buoyancy results in a fast rise of hot partially molten plumes with the porosity of a few percent that extracts most of the liquid present at the silicates interface. For certain parameters range, the liquid in the plumes stays liquid,

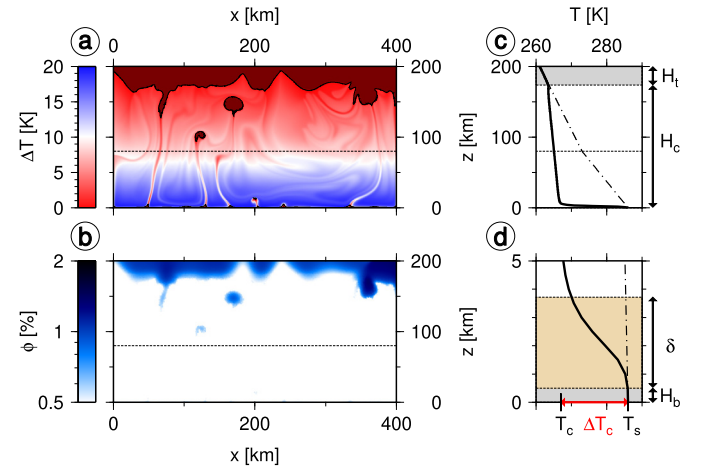


Fig. 2. Results of the reference simulation. (a) Temperature difference from the melting temperature $\Delta T = T_m - T$. The dark red indicates the temperate (partially molten) ice and the thin horizontal line shows the phase change between ice VI (bottom 80 km) and ice V (top 120 km). (b) Porosity (water volume fraction in the mixture). (c) Average temperature profile (solid) and melting temperature (dot dashed). Gray region marks the top temperate ice layer. Thickness of the different layers is indicated (top temperate layer H_t , convective layer H_c). (d) As in (c) but only bottom 5 km. Gray region indicates the bottom temperate layer H_b and brown region the hot thermal boundary layer (TBL) δ . Temperature contrast across the hot TBL, $\Delta T_c = T_s - T_c$, is indicated by red arrow (cf. text for more details).

in other cases, it freezes during its ascent but melts again in the top temperate layer (cf. below). Let us note that the hot thermal boundary layer (TBL, δ , brown area in panel d) is part of the convective layer H_c . Varying the model parameters significantly varies the convection pattern as well as the conditions at the silicates interface, specifically the presence or not of melt. We summarize the results of all performed simulations in Table 2 and we also provide the steady-state temperature profiles for all simulations in Figs. 3, S2–S6.

Generation of melt at the silicates interface substantially facilitates the exchange of volatiles between the silicates and the ocean since volatiles can be dissolved in water. For example, as much as 1.6×10^3 μmol of argon can be dissolved in 1 kg of water at $P = 1$ bar and $T = 286$ K (Hamme and Emerson, 2004). This is the melting temperature at Titan's silicates/HP ice interface, which makes the above solubility value a conservative estimate since the pressure at the silicates/HP ice interface is much larger

Table 2

List of simulations and their outputs.

H (km)	q_s (mW/m ²)	μ_0^V (Pa s)	Q (W/kg)	T_s (K)	H_b (km)	Q_m^0 (mW/m ²)	δ (km)	q_δ (mW/m ²)	T_c (K)	v_i^{rms} (m/yr)	H_t (km)	Φ_0 (m/kyr)	Ra	reg #
50	20	10^{15}	0	286	2.0	15.8	2.6	2.8	281	0.50	18.0	1.72	8.2×10^4	1
100	5	10^{15}	0	285	0.0	0.0	3.6	4.8	275	0.23	20.5	0.42	3.3×10^6	4
100	10	10^{15}	0	286	1.0	1.4	2.9	6.7	275	0.36	21.5	0.89	3.3×10^6	2
100	20	10^{15}	0	286	1.5	16.6	3.3	6.7	275	0.61	23.0	1.71	3.0×10^6	2
200	5	10^{15}	0	279	0.0	0.0	4.0	5.0	267	0.30	21.5	0.44	4.2×10^7	4
200	10	10^{15}	0	286	0.0	0.8	3.6	8.5	267	0.41	23.5	0.80	6.1×10^7	3
200	20	10^{15}	0	286	0.5	16.7	3.2	11.6	267	0.75	26.5	2.11	5.8×10^7	2
200	40	10^{15}	0	286	1.5	34.3	3.0	12.3	267	1.72	29.0	3.71	5.4×10^7	1
300	5	10^{15}	0	276	0.0	0.0	4.4	5.3	261	0.50	51.8	0.47	1.2×10^8	4
300	10	10^{15}	0	284	0.0	0.0	3.6	10.0	261	0.60	52.5	0.98	1.9×10^8	4
300	20	10^{15}	0	286	0.0	3.0	3.6	11.5	261	0.89	54.0	1.25	2.1×10^8	3
200	20	10^{16}	0	286	0.5	15.6	3.0	10.6	268	0.36	40.0	2.00	4.3×10^7	2
200	20	10^{17}	0	286	0.5	16.2	2.9	10.6	270	0.28	60.0	2.48	2.6×10^7	2
200	10	10^{15}	5×10^{-11}	286	0.0	1.8	3.6	8.4	267	0.63	27.0	1.96	5.7×10^7	3
200	10	10^{15}	10^{-10}	286	0.0	2.0	3.7	8.3	267	0.79	29.0	3.13	5.5×10^7	3

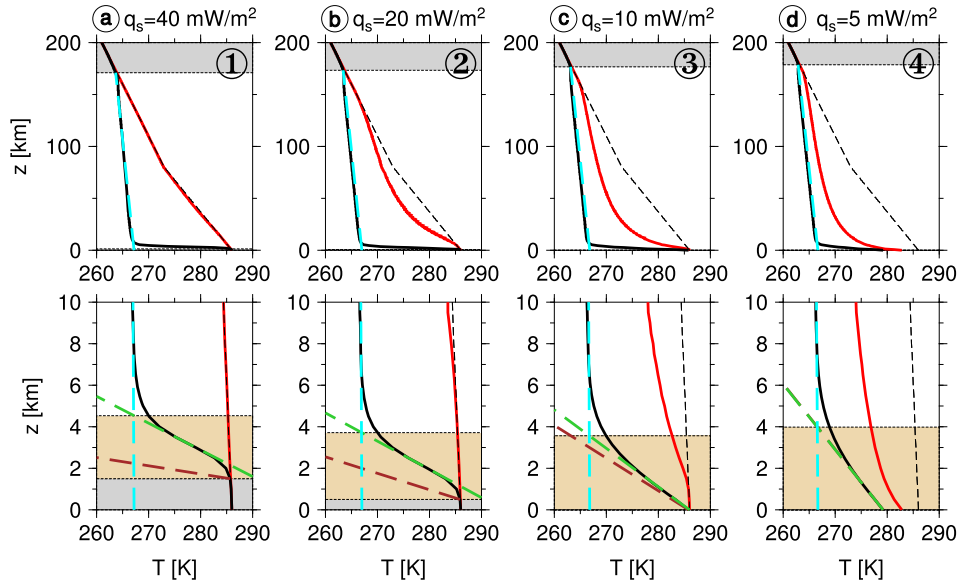


Fig. 3. Temperature profiles for the simulations with $H = 200$ km and $q_s = 40$ (column a), 20 (b), 10 (c), and 5 mW/m² (d). *Top*: maximum (red), average (black), adiabatic (cyan dashed), and melting (black dashed) temperature, top temperate layer (gray rectangles). Circled numbers indicate the particular exchange regime. *Bottom*: detail of the bottom 10 km – temperature gradient corresponding to the prescribed heat flux from silicates (brown dashed), TBL temperature gradient (green dashed), hot TBL (brown rectangles), and bottom temperate layer (gray rectangles).

($P_s = 790$ MPa, cf. Table A.3) and solubility of gases increases with pressure. The dissolved volatiles can be transferred through the convecting layer and then released into the ocean. This process is more efficient than if volatiles would have to be captured and advected by the convecting solid ice.

Based on the steady-state temperature profiles, we have identified four regimes with different degree of material exchange between the silicates and the ocean that characterize the possibility of volatiles exchange through the HP ice layer (Fig. 3):

1. **direct exchange** (column a): melting occurs at the bottom (silicates) and top (ocean) boundaries (H_b , H_t) as well as in the upwelling hot plumes: $T_{\text{max}} = T_m$ (red line)
2. **indirect exchange** (column b): melting occurs at the bottom and top boundaries (H_b , H_t) but melt freezes in the upwelling hot plumes ($T_{\text{max}} < T_m$)
3. **limited exchange** (column c): melting occurs at the top boundary (H_t), a small amount of melt may be transiently

produced at the bottom boundary, but the temperate layer is not identified ($H_b \rightarrow 0$)

4. **no exchange** (column d): melting occurs at the top boundary (H_t) but the temperature at the silicates interface is below the melting temperature ($T_s < T_{ms}$) and thus no melt is produced at the bottom

We now investigate the processes characteristic for each domain and discuss the effects of relevant model parameters. We start at the ocean interface (H_t) and then move downwards through the convective domain (H_c) to the silicates interface (H_b). We focus on deriving scaling laws that would allow us to predict the occurrence of melting at the silicates interface as well as describe the thermal profile within the HP ice layer and the flux of water into the ocean. These laws are derived with the aim of being used in a one-dimensional thermal evolution model where the silicates heat flux q_s and the HP ice layer thickness H will be part of the solution while HP ice viscosity and other material parameters will be kept as unknown model parameters. This demand deter-

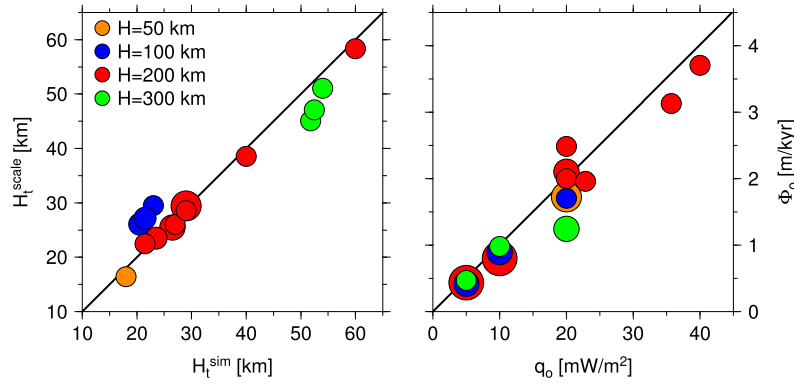


Fig. 4. Left: Top temperate layer thickness H_t^{scale} evaluated using Eq. (2) as a function of H_t^{sim} inferred from the results of numerical simulations. Different colors correspond to different HP ice layer thicknesses H . Right: The water flux at the HP ice/ocean interface Φ_o as a function of total heat flux q_o (cf. text) for different HP ice layer thicknesses H (colors). The black line shows the scaling Eq. (3).

mines the particular form of the scaling laws. Note that all scaling laws derived in the following sections are based on the results of two-dimensional models performed in Cartesian geometry.

3.1. Ocean interface

In all simulations, a layer of temperate ice forms at the ocean interface. Our results (summarized in Table 2) indicate that its thickness H_t depends linearly on the silicates heat flux q_s and scales with the melting point viscosity μ_o of the particular ice phase that forms the top temperate layer, i.e. ice VI for $H = 50$ km, ice V for $H = 100$ and 200 km, and ice III for $H = 300$ km. Moreover, the change in the solid ice phase significantly modulates the density difference $\Delta\rho$ between the solid and liquid phase (cf. right panel of Fig. 1) with the largest values in ice VI (~ 150 kg/m³) and smallest in ice III (~ 35 kg/m³). Smaller density difference leads to smaller water velocities and thus water extraction is less efficient for $H = 300$ km and more efficient for $H = 50$ km. For the limit case of $\Delta\rho \rightarrow 0$ there would be no extraction as water would not be buoyant with respect to ice. To take this effect into account, we scale the top temperate layer thickness by the expected density difference at the ocean interface, $\Delta\rho_o = \rho_i|_{z=H} - \rho_w|_{z=H}$. Finally, the inclusion of internal heating Q also increases the top temperate layer thickness as it triggers more melting within the whole HP ice layer. To include its effect into the scaling, we sum two heat fluxes – the heat coming from the silicates, q_s , and the heat generated by tidal dissipation within the HP ice layer, $\sim \rho_i Q H$. We thus obtain the total heat flux through the top boundary if there was no melting within the HP ice layer:

$$q_o = q_s + \rho_i Q H \quad (1)$$

For cases without internal heating clearly $q_o = q_s$. By combining all dependencies together, we obtained the following scaling law:

$$H_t = (0.06 q_o + 6.5) \mu_o^{0.18} \Delta\rho_o^{-1.1} \quad (2)$$

that gives the thickness H_t in kilometers with q_o in mW/m², μ_o in Pa s, and $\Delta\rho_o$ in kg/m³. The left panel of Fig. 4 shows the scaled thickness H_t^{scale} evaluated using Eq. (2) as a function of H_t^{sim} inferred from the numerical simulations. Different colors correspond to different HP ice layer thickness H . The scaling Eq. (2) gives a very good estimate of H_t in most of the cases. For HP ice layer thicknesses $H = 100$ and 300 km (blue and green), the relative error is less than 30%.

In steady-state, the amount of water generated within the HP ice layer, M_{gen} , must equal the amount of water flowing into the ocean, $M_{\text{out}} = \Phi_o \rho_w S t$ with $\Phi_o = \phi v_w$ the volumetric water flux

through the HP ice/ocean interface (v_w is the outflow water velocity), S its area, and t the time. In cases without internal heating, the amount of generated water is governed by the incoming heat flux from the silicates, i.e. $M_{\text{gen}} = q_s S t / L$ with L the latent heat of melting, while if there is some heating in the volume, it leads to more melting. Using Eq. (1) for the total heat flux, we arrive at the following scaling law for the water flux into the ocean:

$$\Phi_o = \frac{q_o}{\rho_w L} \quad (3)$$

The right panel of Fig. 4 shows the water flux Φ_o inferred from the numerical simulations as a function of q_o . The black line shows the scaling Eq. (3) and indicates that it provides a good estimate of the water flux Φ_o through the HP ice/ocean interface for all simulations.

3.2. Convective layer

The inclusion of transition between the solid ice phases that occurs within the convective layer affects several quantities. First, it modulates the ice density ρ_i and thus also the density difference $\Delta\rho$, which mostly influences the extraction of melt – we have discussed this effect in the previous section. Change of solid phase may also lead to change in viscosity. Ice V was found to be possibly harder to deform than ice VI (Sotin and Poirier, 1987) and thus we have varied the viscosity prefactor μ_o^V in ice V over two orders of magnitude (10^{15} – 10^{17} Pa s) while keeping the same value (10^{15} Pa s) for ice VI. Our results indicate that varying the viscosity in ice V has negligible effect on the heat transfer from the silicates (which is determined by the viscosity of the bottom layer) and thus on the occurrence of melt at the silicates interface (cf. the next section). On the other hand, it substantially affects the top temperate layer thickness (cf. the previous section) as well as the convective velocities. Fig. 5 depicts the temperature difference from T_m (left) and the ice velocity (right) for the three simulations with different ice V viscosity and shows a significant change in the convection pattern. For $\mu_o^V = 10^{15}$ Pa s (reference case, no viscosity jump between ice VI and ice V, top row), convection is chaotic. Increase in ice V viscosity by one order (middle row) leads to stabilization of the large scale pattern. The smaller spatial scale upwellings driven by density instabilities due to the presence of melt are found within ice VI and ice V although their velocity is reduced in ice V. Increasing further the ice V viscosity to $\mu_o^V = 10^{17}$ Pa s (bottom row) results in formation of a large scale flow pattern within the computational domain. Smaller scale features in ice VI are advected towards the right boundary by the global flow and are not present in ice V. The change in size of the convective cell is in agreement with the linear stability analysis

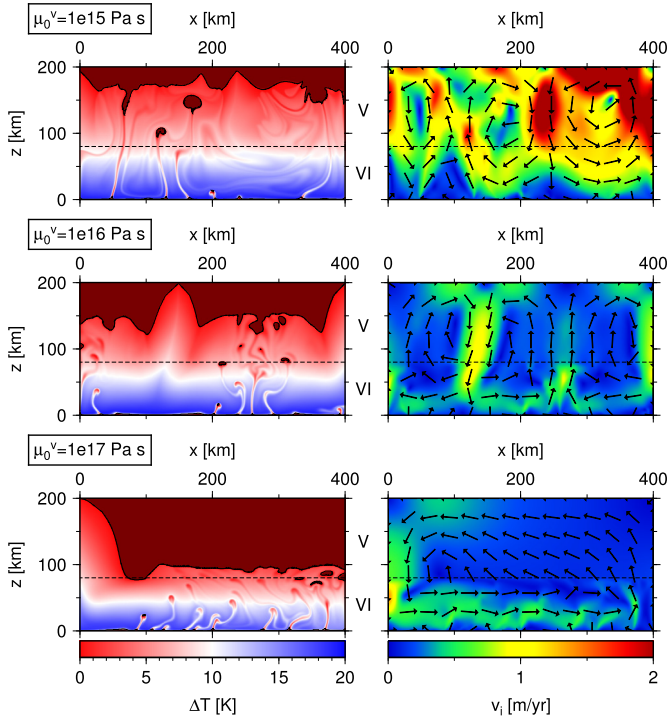


Fig. 5. Effect of jump in reference viscosity μ_0 between ice VI (fixed for all simulations) and ice V: $\mu_0^V = \mu_0^{VI}$ (top), $\mu_0^V = 10 \mu_0^{VI}$ (middle), and $\mu_0^V = 100 \mu_0^{VI}$ (bottom). Horizontal dashed lines indicate the solid phase change with the particular HP ice phases denoted by Roman numerals. *Left:* Temperature difference from the melting temperature ($\Delta T = T_m - T$). Dark red color indicates the temperature ice. *Right:* Ice velocity v_i . Arrows show direction only, the amplitude is indicated by color.

performed for two fluids of different viscosity, density and thickness (Sotin and Parmentier, 1989).

In our simulations, we do not observe any cold thermal boundary layer (TBL) at the interface with the ocean and convection is therefore characterized by hot upwelling plumes. The root-mean-square (rms) velocities of both ice and water are between 0.2 and 2 m/yr giving the travel times from the interface with the silicates to the interface with the ocean (computed as H/v_i^{rms}) on the order of 0.1 to 1 Myr. The longest times correspond to simulations where melting does not occur at the silicates interface (regime #4) and thus there is no additional buoyancy due to the presence of melt. On the other hand, with more melt being generated at the silicates interface (regimes #2 and 1), melt buoyancy significantly enhances the convective transport.

3.3. Silicates interface

The occurrence of melt at the interface with the silicate core is governed by the efficiency of solid state convection within the HP ice layer which is characterized by the Rayleigh number defined as:

$$\text{Ra} = \frac{\alpha_i \rho_i^2 g c_i^p \Delta T_c H_c^3}{k_i \mu_0^{VI}} \quad (4)$$

The material parameters are listed in Table 1, H_c is the thickness of the convective layer, and ΔT_c is the temperature contrast across the hot thermal boundary layer (TBL):

$$\Delta T_c = T_s - T_c \quad (5)$$

with T_s the temperature at the silicates interface and T_c the temperature at the top of the hot TBL (cf. Fig. 2d). For small Rayleigh numbers, the convective heat transfer is not very efficient, the ice

temperature at the silicates interface reaches the melting temperature, and melting occurs. On the other hand, for large Rayleigh numbers, all the heat coming from the radiogenic decay in the core is efficiently transferred through the ice layer by convection and the ice temperature at the silicates interface does not reach the melting point.

To characterize the thermal state of this interface when there is no melt ($T_s < T_{ms}$, regime #4), we have performed thermal boundary layer analysis that relates the nondimensional thickness of the hot thermal boundary layer

$$\delta' = \frac{\delta}{H_c} \quad (6)$$

with the Rayleigh number Ra . The particular form of the scaling law depends on the prescribed boundary conditions. For no slip, Mitrović and Jarvis (1987) found:

$$\delta'_{\text{ns}} = 0.630 \text{Ra}^{-0.159} \quad (7)$$

while for free slip, the data from Jarvis (1984) give:

$$\delta'_{\text{fs}} = 1.108 \text{Ra}^{-0.303} \quad (8)$$

Using the results of our simulations with no melting at the bottom interface (regime #4), we obtained the following scaling for the non-dimensional hot TBL thickness:

$$\delta' = 2.705 \text{Ra}^{-0.273} \quad (9)$$

Fig. 6 (left) shows the values of δ' from the four simulations in regime #4 as functions of Rayleigh number. The black line shows the obtained scaling law (Eq. (9)), while the gray lines indicate the published scaling laws for no slip (Eq. (7), full line) and free slip (Eq. (8), dashed line). As expected for our setting with no slip condition at the bottom and free slip condition at the top boundary (cf. Appendix B), the obtained power-law coefficient of -0.273 is between the value of -0.159 when both boundaries are no slip and -0.303 when both boundaries are free slip. The cyan line corresponds to the same scaling law obtained for Ganymede (Kalousová and Sotin, 2018) and indicates a good agreement between the two equations.

If melting does not occur at the silicates interface, the TBL heat flux q_δ equals the silicates heat flux q_s and we can write

$$q_\delta = \frac{k_i \Delta T_c}{\delta} = q_s \quad (10)$$

To obtain the critical Rayleigh number to reach the melting temperature at the interface with the silicates, we proceed as follows:

1. we combine Eqs. (6), (9), and (10)
2. we assume that the temperature at the silicates interface approaches the melting temperature from below ($T_s \nearrow T_{ms}$)

By this procedure, we arrive at:

$$\text{Ra}_c = \left[\frac{k_i (T_{ms} - T_c)}{2.705 H_c q_s} \right]^{-\frac{1}{0.273}} \quad (11)$$

Apart from the incoming silicates heat flux q_s , this value depends on the convective layer thickness H_c and the temperature T_c – the results of numerical simulations. Note that Ra_c does not explicitly depend on ice viscosity. To predict the occurrence or not of bottom melting without the necessity to perform the simulations, we use the following procedure:

1. we evaluate the thickness H_t of the top temperate layer for a range of H (which gives the density difference at the ocean interface, $\Delta \rho_o$) and q_s using the scaling Eq. (2)

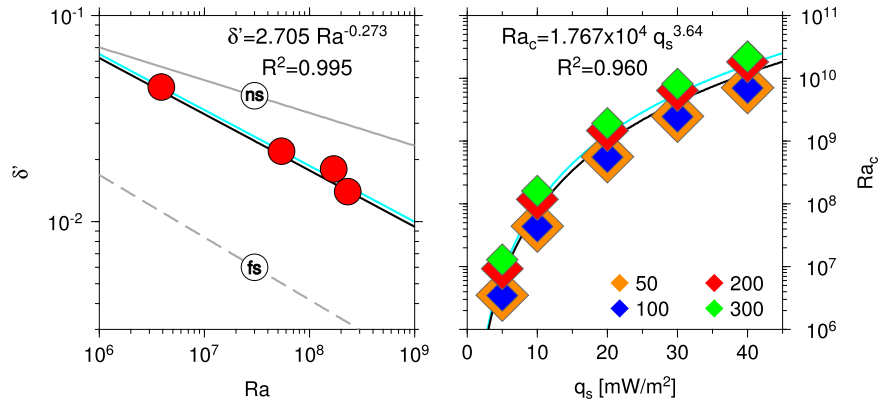


Fig. 6. Left: Nondimensional thickness of the hot TBL (δ') as a function of Rayleigh number Ra for simulations in regime #4 (red circles). Black line shows the power-law scaling (Eq. (9)), gray lines indicate scaling laws from Mitrovia and Jarvis (1987) (no slip, Eq. (7), full line) and Jarvis (1984) (free slip, Eq. (8), dashed line), and cyan line shows the scaling law found in Kalousová and Sotin (2018) for Ganymede (Eq. 4 in that paper). Right: Critical Rayleigh number for bottom melting Ra_c computed for thicknesses $H = 50$ – 300 km (colors) and silicate heat fluxes $q_s = 5$ – 40 mW/m² (x-coordinate). Black line shows the power-law scaling (Eq. (12)), cyan line shows the corresponding scaling law found in Kalousová and Sotin (2018) for Ganymede (Eq. 7 in that paper).

2. we neglect the thickness of the bottom temperate layer H_b (see below) and write $H_c = H - H_t$
3. we estimate the temperature T_c at the top of the hot TBL assuming that the average temperature in the convective layer follows the adiabatic gradient, i.e. $T_c = T_{m|z=H_c}(1 + \alpha_i g H_c / c_i^p)$ (cyan dashed lines in Fig. 3)
4. using these estimates for H_c and T_c , we evaluate Ra_c from Eq. (11)

Note that we use the same viscosity for all ice phases and only consider the reference value of 10^{15} Pa s which is possible due to the non-dependence of the critical value of Ra_c evaluated from Eq. (11) on ice viscosity. Right panel of Fig. 6 shows the obtained values of Ra_c as a function of silicates heat flux q_s for different HP ice layer thickness H (color). Observing that the values of Ra_c are mainly dependent on q_s , we performed a power-law scaling to obtain

$$Ra_c = 1.767 \times 10^4 q_s^{3.64} \quad (12)$$

which is shown by the black line. This scaling is again in good agreement with the Ganymede scaling law (cyan line) and thus the presence of different solid ice phases does not significantly modify this scaling. If the Rayleigh number defined by Eq. (4) exceeds the critical value Ra_c defined by Eq. (12), convection becomes too efficient, temperature at the silicates interface will not reach the melting temperature, and melting will not occur at the silicates interface.

With the aim of developing a one-dimensional thermal evolution model, we will now derive a scaling for the critical silicates heat flux for bottom melting q_s^c as a function of HP ice layer thickness H and ice viscosity μ_0^V which is straightforward to implement into the evolution model. Note that we assume that ice VI and ice V have the same viscosity ($\mu_0^V = \mu_0^I$). For the values of viscosity prefactor μ_0^V between 10^{13} and 10^{17} Pa s (cf. Appendix C for discussion on this range), range of silicates heat flux q_s and HP ice layer thickness H , we successively evaluate: (i) H_t from Eq. (2), (ii) $H_c = H - H_t$, (iii) $T_c = T_{m|z=H_c}(1 + \alpha_i g H_c / c_i^p)$, (iv) ΔT_c from Eq. (5), and (v) Ra from Eq. (4). Comparison of these values with Ra_c (Eq. (12)) evaluated for the corresponding range of q_s provides us with the critical silicates heat flux for bottom melting q_s^c as a function of the HP ice layer thickness H and the ice VI viscosity μ_0^V . The obtained values are plotted in the left panel of Fig. 7 (light colors) and confirm that the smaller the viscosity (leading to more efficient heat transfer by solid state convection), the larger the heat flux that is necessary for bottom melting for a given HP

ice layer thickness and vice versa. Let us note that due to density jumps between the different HP ice phases (at $H = 80$ km for the ice VI–ice V transition and at $H = 250$ km for the ice V–ice III transition, cf. also top of the left panel), the critical heat flux also contains jumps at the corresponding thickness H .

Performing a linear scaling of this functional dependence with respect to H and then scaling the obtained coefficients with respect to μ_0^V , we arrive at the following expression for the critical heat flux for bottom melting (in mW/m²):

$$q_s^c = 1.044 \times 10^3 (\mu_0^V)^{-0.29} H - 0.45 \quad (13)$$

with μ_0^V in Pa s and H in km. This scaling is shown in dark colors in Fig. 7 (left). Varying other model parameters (ice V viscosity μ_0^V and internal heating Q) has negligible to no effect on the occurrence of melt at the silicates interface as these parameters affect mainly the top part of the layer (μ_0^V) or the whole HP ice layer (Q). The implications for Titan's evolution are discussed in the next section.

We now investigate the amount of melt that can be produced at the silicates interface when melting occurs in the bottom temperate layer (regimes #1–2) with the thickness H_b between 0.5 and 2 km. Let us define the heat consumed on melting at the silicates interface as

$$Q_m^0 = \frac{L}{2H} \int_0^{2H} \int_0^{z_0} \Gamma dx dz \quad (14)$$

where z_0 is the height of the bottom element and Γ is the melting rate (cf. Appendix B). In simulations where melting occurs in the bottom temperate layer H_b (regimes #1–2), all the supplied heat is consumed on melting: $Q_m^0 \sim q_s$ (cf. Table 2). This is because temperature in the temperate layer equals the melting temperature and we assume that the phases are in thermal equilibrium (i.e. the mixture temperature cannot exceed the melting temperature). This allows us to estimate the melting rates as

$$\Gamma = 4\pi R_s^2 \frac{q_s}{L} \quad (15)$$

with R_s the radius of the silicate core. We do not observe the accumulation of large amounts of melt at the silicates interface in our results as most of generated water is very efficiently transferred by the upwelling hot plumes. This is important as our models do not predict the formation of water pools trapped between the HP ice and the silicates similar to the terrestrial deep Antarctic lakes,

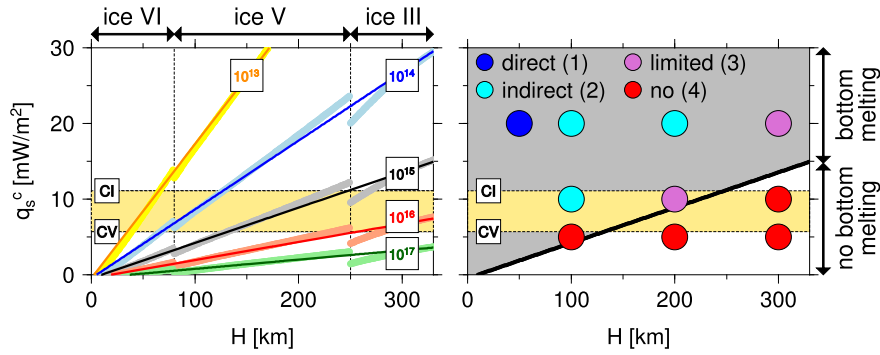


Fig. 7. Left: Critical heat flux q_s^c for bottom melting as a function of HP ice layer thickness H for different ice VI viscosities $\mu_0^{VI} = 10^{13}$ – 10^{17} Pa s (colors): values obtained from comparison of Ra (Eq. (4)) and Ra_c (Eq. (12)) (light colors) and the scaling Eq. (13) (dark colors). The jumps in light color curves (marked also by vertical dotted lines) correspond to solid phase change between ice VI and ice V ($H = 80$ km) and ice V and ice III ($H = 250$ km). The light yellow rectangle represents the estimated present values of q_s for two end-member models (CI and CV elementary composition of the core, cf. Section 4 for more details). Right: Results of numerical simulations (colored circles) as function of model parameters H and q_s . Colors indicate the particular exchange regime (cf. text). The black line shows the scaling law for critical silicates heat flux q_s^c (Eq. (13)) assuming $\mu_0^{VI} = 10^{15}$ Pa s (cf. also left panel). The gray region shows the couples of (H, q_s) where bottom melting occurs while the white region corresponds to no bottom melting setting. The light yellow rectangle is the same as in the left panel.

such as lake Vostok, trapped between the ice and the silicate crust. Such lakes would have been interesting habitats for potential development of life.

4. Discussion

In this study, we have shown that melt can form at the silicates/HP ice interface for large values of the heat flux q_s and small layer thickness H (Fig. 7, left). The right panel of Fig. 7 shows the results of our two-dimensional simulations in terms of the particular exchange regime (colors) as a function of HP ice layer thickness H and silicates heat flux q_s for viscosity $\mu_0 = 10^{15}$ Pa s. The blue colors (regime #1 and 2) correspond to substantial melting at the silicates interface and red (regime #4) indicates no melting. Violet (regime #3) represents a limited exchange regime where bottom melting is occurring transiently. The solid black line shows the critical heat flux q_s^c given by equation (13): if the incoming silicates heat flux q_s is larger than the critical value q_s^c for a given HP ice layer thickness H , melting occurs at the silicates interface (gray region). For smaller heat fluxes, the temperature at the silicates interface is below the melting temperature and no melting occurs (white region). The scaling law (black line) explains very well the results of numerical simulations (colored circles).

The present day value of the heat flux coming out of the silicate core can be bounded between 5.7 and 11.1 mW/m² (light yellow rectangle in Fig. 7). These limits encompass models of heat transfer by conduction and convection for both CV and CI-type elementary composition (calculations are detailed in the Supplementary material). For the reference HP ice viscosity of 10^{15} Pa s, this study predicts that melt could still be produced at the silicates interface at the present time if the HP ice layer thickness is less than 248 km or 132 km for the CI or CV model, respectively (Fig. 7). For smaller viscosity of 10^{14} Pa s, our scaling law predicts that melt could be produced at the silicates interface only if the HP ice layer thickness is less than 127 km or 68 km, respectively. On the other hand, for larger viscosity of 10^{16} Pa s, we predict bottom melting for layer thickness less than 257 km for the CV model. For the CI model, melt is produced at the silicates interface independently of the HP ice layer thickness. Although the HP ice viscosity is poorly constrained (Kalousová et al., 2018) and would benefit from new laboratory experiments, its range can be further narrowed by taking into account the orbital evolution since viscosity will strongly influence the amount of tidal dissipation that will in turn influence the orbital evolution. These effects will be assessed in a future study. Even though the Cassini observations did not provide the information on the thickness of the HP ice layer, fu-

ture missions such as the recently selected mission Dragonfly that will carry seismometers could provide such information (Lorenz et al., 2019).

Following Baland et al. (2014) who predict the ice I crust thickness smaller than 100 km and assuming pure H₂O hydrosphere, the present HP ice layer thickness is less than 140 km (determined by the ocean adiabat, cf. model 2 in Table A.3). For this value, melt is predicted at the silicates interface even for CV models if the ice VI viscosity at the melting point is 10^{15} Pa s or larger (Fig. 7).

The evidence for outgassing comes from the presence of ⁴⁰Ar in Titan's atmosphere (Niemann et al., 2005; Waite et al., 2005; Atreya et al., 2006) because ⁴⁰Ar is produced by the decay of ⁴⁰K that is initially contained in the silicate fraction. Recently, Miller et al. (2019) proposed that up to 50% of Titan's atmospheric nitrogen comes from the decomposition of IOM in the silicate core. Since both argon and nitrogen are soluble in water, its presence at the silicates interface and its subsequent transfer to the ocean by processes described in the present study provides an easy means to deliver these volatiles from the silicates to the ocean. Titan's silicate interior is likely hydrated to account for its low density (Castillo-Rogez and Lunine, 2010; Néri et al., 2020). Choosing the CI model for the K/Si ratio, the present production rate of argon is 1.6×10^{13} mol/Myr. The melting rate of ice (production rate of water) at the silicates interface is given by Eq. (15). Assuming the CI heat flux and the parameters of this study, it is equal to 6.2×10^{19} kg/Myr. The ratio of these two numbers gives the amount of argon that would need to be dissolved. It is equal to 0.26 μ mol of argon per kg of water which is more than 3 orders of magnitude lower than the solubility of argon in water (Hamme and Emerson, 2004). Therefore all potential argon, if it can diffuse to the silicates/HP ice interface, can be dissolved in water and thus the HP ice layer does not represent a barrier for its transfer to the ocean as long as the conditions are fulfilled for melt being produced at the interface (Fig. 7). Similar calculations can be done for nitrogen. However, Miller et al. (2019) do not provide a quantitative production rate of nitrogen which must be related to the thermal evolution of Titan's core. Once such a model is constructed, then the scaling laws described in the present study can be used to predict the transfer of nitrogen to the ocean.

Models simulating the chemical evolution of Titan's silicate core (Castillo-Rogez and Lunine, 2010; Néri et al., 2020) suggest that the silicates dehydrated during the thermal evolution. It is then possible that the water resulting from the dehydration migrated to the interface with the HP ice layer bringing with it salts and volatiles leached during the migration. A quantitative model describing the dehydration does not exist at present time. Once avail-

able, it will be possible to study the impact of salty water flux at the silicates/HP ice interface and investigate the implications for the transport of salts to the ocean.

Ganymede and Titan are very similar in mass and radius but, as indicated by the different values of Mol, their interior exhibits different characteristics. Titan's HP ice layer is much thinner than Ganymede's (Néri et al., 2020). A first consequence of a thinner HP ice layer on Titan is that melting is more likely than on Ganymede. Another consequence is that the thickness of Titan's ice VI layer is small (~80 km) and the transition to ice V has to be taken into account. We show that the ice VI-ice V transition does not fundamentally modify the main characteristics of the HP ice layer in terms of the presence of temperate layers. Callisto has also similar radius and mass and, based on the measurements of an induced magnetic field, Khurana et al. (1998) proposed that it harbors a deep ocean. However, a recent study by Hartkorn and Saur (2017) shows that the ionosphere could produce similar signals and suggests that the ocean is much deeper or does not exist at all. Callisto is likely much less differentiated than Titan (and Ganymede) based on the value of its Mol inferred from the measurements of the degree 2 gravity coefficients (Anderson et al., 2001). However, the value of Mol assumes hydrostatic equilibrium which may not be valid since the ratio of the two gravity coefficients differs from the hydrostatic state (Anderson et al., 2001). As shown by Gao and Stevenson (2013), non-hydrostatic forces can severely affect the value of the Mol. Application of the present model to Callisto will await a more precise knowledge of the induced magnetic field and the gravity coefficients by the JUICE mission (Grasset et al., 2013).

5. Conclusions

Using a numerical code that solves the problem of two-phase thermal convection of solid ice and liquid water mixture, we investigated the dynamics of Titan's HP ice layer. Our results indicate that the occurrence of melt at the interface with the silicate core depends on the ice viscosity, the HP ice layer thickness and the heat flux coming from the silicate core. We found a scaling law that gives the critical value of silicates heat flux above which melting occurs as a function of the HP ice layer thickness and the ice VI viscosity.

If melting occurs at the silicates interface, argon, nitrogen and other volatiles coming out of the silicates could be dissolved in water and advected through the HP ice (liquid or frozen) on timescales between 0.1 and 1 Myr. After reaching the temperate ice layer at the ocean interface, they would be extracted into the ocean. We provide scaling laws for both, the top temperate layer thickness and the water flux into the ocean.

Based on Cassini observations and reasonable values of ice VI viscosity, our results suggest that exchange processes through Titan's deep HP ice layer might still be ongoing. A thermal evolution model that will use the scaling laws derived in the present study will help predict the timing of exchange processes between Titan's interior and its atmosphere.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Titan's interior structure model

Titan's interior structure is determined following the approach described by Vance et al. (2014) for Ganymede, using the values of the total mass ($M = 1.3453 \times 10^{23}$ kg) and the moment of inertia ($C = 0.341$) of Titan (Durante et al., 2019). Another strong constraint comes from the presence of a global ocean inferred from several independent measurements. Assuming a 1D spherical model, the total mass can be calculated as a sum of masses of nested spherical shells with radius r , thickness dr and density $\rho(r)$ (constant within the given shell):

$$M = \int dm = 4\pi \int_0^{R_s} \rho(r)r^2 dr + 4\pi \int_{R_s}^R \rho(r)r^2 dr = \frac{4}{3}\rho_s R_s^3 + M_{H_2O} \quad (A.1)$$

Here $R = 2574.706$ km is the radius of Titan, R_s and ρ_s are the unknown radius of the silicate core and its density which is considered constant, and M_{H_2O} includes the mass of the low-pressure ice I crust, the ocean, and the high-pressure ice layer. Similarly, the moment of inertia can be calculated from:

$$C = \int x^2 dm = \frac{8}{3}\pi \int_0^{R_s} \rho(r)r^4 dr + \frac{8}{3}\pi \int_{R_s}^R \rho(r)r^4 dr = \frac{8}{15}\pi \rho_s R_s^5 + C_{H_2O} \quad (A.2)$$

where x is the distance of any element of mass (dm) from the spin axis and C_{H_2O} is the moment of inertia of the hydrosphere. Equations (A.1) and (A.2) are solved iteratively to obtain the two unknowns, density (ρ_s) of the silicate core and its radius (R_s). Note that the model could be improved by using equations of state of the silicates but these depend on the type of silicates. This problem has recently been addressed by Néri et al. (2020) and is out of the scope of the present paper.

The density within the hydrosphere is determined using the equations of state of water and the different phases of ice which requires the knowledge of temperature and pressure. We assume that the temperature follows the adiabatic profile in the ocean and the melting curve within the HP ices. Note that the temperature profiles found in this study do not modify the calculation of the Mol as the temperature profile is close to the melting curve. The pressure profile is determined after a few iterations since density and gravity acceleration are necessary to calculate the pressure. The gravity in the H_2O layer increases from 1.354 m/s^2 at the surface to 1.500 m/s^2 at the silicates/HP ice interface.

As in the case of Ganymede (Vance et al., 2014), we have evaluated the interior structure for four different values of the temperature at the base of the ice I layer ($T_1 = 255\text{--}270$ K). We obtained the corresponding pressure from the assumption that this interface is at the melting point and we anchored the density profiles along the melting curve of ice I. In the obtained models, we found

Table A.3Interior structure models for different temperatures at the ice I/ocean interface T_1 (bold).

Symbol	Physical property	Layer/Interface	Unit	Model 1	Model 2	Model 3	Model 4
H_1	thickness	ice I	km	141	110	74	33
T_1	temperature	ice I/ocean	K	255	260	265	270
P_1	pressure	ice I/ocean	MPa	182.1	141.6	94.9	42.2
H_o	thickness	ocean	km	121	246	391	465
H	thickness	HP ice	km	239	139	28	0
H_{H_2O}	thickness	hydrosphere	km	501	495	493	498
T_s	temperature	core/HP ice	K	286.9	285.7	284.5	289.0
P_s	pressure	core/HP ice	MPa	819.9	802.0	783.8	790.6
g_s	gravity	core/HP ice	m/s ²	1.488	1.491	1.496	1.500
R_s	radius	silicate core	km	2074.4	2079.6	2082.1	2077.0
ρ_s	density	silicate core	kg/m ³	2568.5	2567.3	2573.1	2581.0

that the interface between the hydrosphere and the silicate core has a radius of $R_s = 2078 \pm 3$ km and is located at a pressure of $P_s = 799 \pm 14$ MPa (cf. Table A.3). These results led us to use rounded values of $P_s = 790$ MPa and $R_s = 2080$ km for the pressure and the radius of the silicates/HP ice interface which are consistent with those proposed by Castillo-Rogez and Lunine (2010) and Fortes (2012).

The obtained value of R_s gives the total hydrosphere thickness of 495 km with the HP ice layer thickness obtained from the model ranging between 0 and 239 km (Table A.3). Here, the absence of HP ice and thus a direct contact of the ocean and silicates is found for $T_1 = 270$ K. However, such a large temperature at the ice I/ocean interface implies a crustal thickness of less than 35 km, which is lower than the current estimates proposed by several studies (Beghin et al., 2012; Baland et al., 2014; Mitri et al., 2014). The maximum thickness of the HP ice layer is on the order of 330 km which corresponds to the triple point between ice I, ice III and water. Since the current thickness is highly uncertain, in the present study, we investigate the dynamics of the HP ice layer for thicknesses between 50 and 300 km. Left panel of Fig. 1 shows the interior structure of Titan within the phase diagram of H₂O. For the HP ice layers thicker than 80 km, the solid phase change from ice VI to ice V has to be taken into account. For cooler hydrospheres with very thick HP ice layers ($H > 250$ km), another phase change from ice V to ice III occurs.

Appendix B. Two-phase convection numerical model

We treat the HP ice layer material as a mixture of two coexisting phases – solid ice (subscript i) and liquid water (subscript w) (Souček et al., 2014). In this approach, the amount of water is described by porosity ϕ that is defined as a volume fraction of water in the whole mixture. In our simulations, porosities are typically on the order of a few percent indicating that ice is the dominant phase in the mixture. We assume the zero compaction length approximation of the full two-phase equations which prevents the generation of porosity waves (cf. Souček et al., 2014, for more details). While affecting the porosity field, this approximation has a small effect on the percolation velocity which is the main characteristic important in our study. In our recent studies tailored to the HP ice layer of Ganymede (Kalousová et al., 2018; Kalousová and Sotin, 2018), we assumed that the HP ice layer was only composed of ice VI. In the case of Titan, the phase change in the solid ice has to be taken into account (Fig. 1). For the sake of simplicity, we assume that the solid-solid phase change only depends on depth (hydrostatic pressure, i.e. we neglect its temperature, dynamic pressure, and time dependence) and thus all phase-dependent quantities are assumed to be also only depth-dependent (Appendix C). Finally, we have also included several new terms in the energy balance neglected in the previous work. The corresponding governing equations are as follows:

$$\nabla \cdot [\phi \mathbf{v}_w + (1 - \phi) \mathbf{v}_i] + \frac{\phi}{\rho_w} \mathbf{v}_w \cdot \nabla \rho_w + \frac{(1 - \phi)}{\rho_i} \mathbf{v}_i \cdot \nabla \rho_i = \frac{\Delta \rho}{\rho_w \rho_i} \Gamma \quad (\text{B.1a})$$

$$\nabla \Pi = -\phi \Delta \rho \mathbf{g} - (1 - \phi) \rho_i \alpha_i (T - T_m) \mathbf{g} + \nabla \left[\frac{1 - \phi}{\phi} \mu_i (\nabla \cdot \mathbf{v}_i) \right] + \nabla \cdot \left[(1 - \phi) \mu_i \left(\nabla \mathbf{v}_i + (\nabla \mathbf{v}_i)^T - \frac{2}{3} (\nabla \cdot \mathbf{v}_i) \mathbf{I} \right) \right] \quad (\text{B.1b})$$

$$\mathbf{v}_w = \mathbf{v}_i - \frac{K_\phi}{\phi \mu_w} (1 - \phi) \Delta \rho \mathbf{g} \quad (\text{B.1c})$$

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_w) + \frac{\phi}{\rho_w} \mathbf{v}_w \cdot \nabla \rho_w = \frac{\Gamma}{\rho_w} \quad (\text{B.1d})$$

$$\rho_i c_i^p \frac{\partial T}{\partial t} + \left[\rho_w \phi c_w^p \mathbf{v}_w + \rho_i (1 - \phi) c_i^p \mathbf{v}_i \right] \cdot \nabla T + \rho_i g T \left[\alpha_w \phi \mathbf{v}_w^z + \alpha_i (1 - \phi) \mathbf{v}_i^z \right] + \Gamma L + \rho_i L_s \mathbf{v}_i \cdot \nabla c = \rho_i Q + \nabla \cdot (k(\phi) \nabla T) + \frac{K_\phi}{\mu_w} (1 - \phi)^2 \Delta \rho^2 g^2 + (1 - \phi) \sigma_i : \nabla \mathbf{v}_i + \frac{1 - \phi}{\phi} \mu_i (\nabla \cdot \mathbf{v}_i)^2 \quad (\text{B.1e})$$

Here ϕ is the porosity (volume fraction of water in the mixture), $\mathbf{v}_w$ and $\mathbf{v}_i$ is the velocity of water and ice, respectively, ρ_w and ρ_i is the water and ice density, respectively, $\Delta \rho = \rho_i - \rho_w$ is the density difference between the two phases, Γ is the melt production rate (positive for melting), $\Pi = P_w - P_i^{\text{ref}}$ is the excess water pressure with respect to the hydrostatic equilibrium pressure of pure ice (P_i^{ref}), $\mathbf{g}$ is the gravity, α_i is the ice thermal expansivity, T is the temperature, T_m is the depth-dependent melting temperature, μ_i is the temperature and depth-dependent ice shear viscosity:

$$\mu_i = \mu_0 \exp \left[\frac{E}{R} \left(\frac{1}{T} - \frac{1}{T_m} \right) \right] \quad (\text{B.2})$$

with μ_0 the melting-point viscosity, E the activation energy, R the universal gas constant; K_ϕ is the porosity-dependent permeability:

$$K_\phi = \begin{cases} K_0 (\phi - \phi_c)^n & \phi \geq \phi_c \\ 0 & \phi < \phi_c \end{cases}, \quad (\text{B.3})$$

with K_0 and n the experimentally determined constants and ϕ_c the percolation threshold; μ_w is the water shear viscosity, t is the time, c_w^p and c_i^p is the isobaric specific heat of water and ice, respectively, α_w is the water thermal expansivity, $\mathbf{v}_w^z$ and $\mathbf{v}_i^z$ is the vertical water and ice velocity, respectively, L is the latent heat of melting of ice, L_s is the latent heat of the solid-solid phase change, c is the solid phase function that indicates the particular solid ice

phase present (i.e. ice VI, ice V, or ice III), Q is the internal heating rate (e.g. due to tides, cf. Appendix C), and $k(\phi)$ is the thermal conductivity of the mixture:

$$k(\phi) = k_i \left(1 - 3\phi \frac{k_i - k_w}{2k_i + k_w} \right) \quad (\text{B.4})$$

with k_i and k_w the thermal conductivity of ice and water, respectively. This expression is derived as a small porosity limit of two-phase mixture conductivity given in McKenzie (1984). All material parameters are listed in Table 1.

Comparison of the above equations with the set of governing equations (1a)–(1e) in Kalousová et al. (2018) shows which physical processes were taken into account here that were neglected in the previous study: (i) depth-dependence of material densities (terms with $\nabla\rho$), (ii) heat advection by water percolation ($\rho_w \phi c_w \mathbf{v}_w \cdot \nabla T$ term in Eq. (B.1e)), (iii) adiabatic heating in both water and ice (third term on lhs of Eq. (B.1e)), (iv) latent heat due to solid-solid phase change (last term on lhs of Eq. (B.1e)), (v) internal heating due to tides (first term on rhs of Eq. (B.1e)), (vi) heating due to relative motion between the solid and liquid phase (third term on rhs of Eq. (B.1e)), (vii) viscous shear heating (fourth term on rhs of Eq. (B.1e)), and (viii) viscous bulk heating (fifth term on rhs of Eq. (B.1e)).

Equations (B.1a)–(B.1e) are solved in a two-dimensional Cartesian box with aspect ratio of 2 and the following boundary conditions. Free slip at the top and side boundaries and no slip at the bottom boundary are considered for the ice velocities. The excess water pressure is determined up to a constant, fixed by setting $\Pi = 0$ in one point of the computational domain. No boundary condition is required for porosity, since the transport equation (B.1d) is hyperbolic and no inflow boundary is present in the current setting. Free water outflow is assumed at the top boundary (interface with the ocean), while the bottom and side boundaries are kept water impermeable. Finally, we fix the temperature at the top boundary to the ocean temperature, we prescribe a fixed heat flux q_s from the silicate core at the bottom boundary, and we keep the side boundaries insulating. The system of governing equations together with these boundary conditions is solved by a Finite Element Method code implemented in the open source FEM software FEniCS (Logg et al., 2012; Alnæs et al., 2015). Details of the numerical method are described in Kalousová et al. (2018). We use a structured mesh with the basic grid of 200×100 (or 300×150 for $H = 300$ km) crossed diagonal triangular elements which is refined at the bottom and top boundaries, as well as at the solid-solid phase change interfaces. Resolution tests were performed in the earlier study (Kalousová et al., 2018).

Appendix C. Material parameters

As mentioned above, solid ice undergoes phase change for the pressure-temperature conditions relevant for Titan's HP ice layer. Therefore, ice VI, ice V, and perhaps even ice III can be present. The values of the different ice VI material parameters come from the same references as in Kalousová et al. (2018) and we also use ice VI values for thermal expansion, thermal conductivity and specific heat of ice V and III. Latent heat of ice VI melting depends on pressure as the slope of the melting curve changes – it varies from 306 kJ/kg at the silicates interface to 265 kJ/kg at the triple point with ice V and we use a linear dependence on vertical coordinate z (in km), L [kJ/kg] = $306 - 0.5125z$ [km]. For ice V, the latent heat varies from 285 to 265 kJ/kg between the two triple points and we choose a constant value of 275 kJ/kg. For ice III, we take the value of 256 kJ/kg (Hobbs, 1974). The latent heat associated with the solid-solid phase change is two orders of magnitude smaller and we use 1 kJ/kg for ice VI-ice V change (Bridgman, 1937) and

4 kJ/kg for ice V-ice III change (Hobbs, 1974). We also use linear dependence on z of densities and melting temperature (Fig. 1). The difference from the measured values is less than 5 kg/m³ and 2 K, respectively.

The heat flux q_s coming from the silicate core varies through Titan's evolution and depends on the amount of radioactive elements and the means of heat transfer (convection/conduction) in the silicate core (Tobie et al., 2006). We consider a plausible range of heat flux values from 5 to 40 mW/m² (cf. Fig. S7).

Permeability is the ability of ice to transmit fluids and depends on numerous parameters. In numerical simulations, a power-law dependence on porosity is mostly used (Eq. (B.3)) with the exponent of 2 or 3 and the threshold porosity ϕ_c on the order of a few percent. For terrestrial sea ice, the value of 5% was reported by Golden et al. (1998) but the higher pressures in the HP ice layer may significantly change this value. Nakano and Fujii (1989) report percolation threshold of about 1% for the silicate rock. We investigated the range of percolation thresholds between 1 and 5% in Kalousová et al. (2018) and found that the larger the threshold value, the thicker the top temperate layer H_t . Increase in the threshold value also leads to warming of the HP ice layer while not significantly modifying the occurrence of melt at the bottom interface. Overall, the effect of threshold porosity is similar to that of viscosity of the particular HP ice phase on the boundary with the ocean (ice V in most of the simulations in the present study). Since we investigate the effect of increase in ice V viscosity, we decided to keep the threshold value as $\phi_c = 1\%$ for the sake of simplicity. We assume $n = 2$ and $K_0 = 10^{-9}$ m² (e.g. Petrich et al., 2006; Golden et al., 2007).

Viscosity is a critical parameter that controls the dynamics of the HP ice layer. For Titan which is less differentiated than Ganymede, the main ice phases are ice VI and ice V. Their viscosity was measured by two groups. Sotin et al. (1985) and Sotin and Poirier (1987) used a sapphire anvil cell to observe the deformation of ice VI and ice V at large values of shear stress. They investigated the variations in a limited range of shear stress and temperature and found a small stress-dependence. For ice VI, the extrapolation of their data to typical convective shear stresses gives viscosities between 10^{13} and 10^{14} Pa s, similar to viscosities of ice I at its melting point which are very close to field measurements on glaciers (cf. Kalousová et al., 2018). Another group led by Durham measured the viscosity of ice V and ice VI in a gas-medium high-pressure deformation rig (Durham et al., 1996, 1997; Durham and Stern, 2001). They found much stronger stress dependence as expected at high values of shear stress. The extrapolation of their experimental data leads to ice VI viscosity several orders of magnitude larger than those extrapolated from Sotin and Poirier (1987). More details on the viscosity of ice VI is given in Kalousová et al. (2018) where we investigated the ice VI viscosity between 10^{14} and 10^{17} Pa s. Here, for the sake of simplicity, we choose the reference value of ice VI viscosity to be $\mu_0^{\text{VI}} = 10^{15}$ Pa s in all numerical simulations. However, we vary this parameter between 10^{13} and 10^{17} Pa s in the derivation of the scaling laws (Section 3.3). We choose an activation energy of 30 kJ/mol for ice VI (Table 1) which was given by Sotin et al. (1985). Durham et al. (1996) report a larger value of 110 kJ/mol which would reduce the temperature difference across the thermal boundary layer that is inversely proportional to the activation energy ($\Delta T \propto RT^2/Q$, Davaille and Jaupart, 1993). However, the effect of a larger activation energy is left to future studies. Note also that we do not take into account viscosity reduction due to the presence of melt – for ice I, De La Chapelle et al. (1999) reported the reduction of viscosity by one order of magnitude for the melt content of 5% and a similar reduction can be expected for the HP ice phases. Although this effect is potentially important and can reduce the impact of the viscosity jump between the different HP ice phases (cf. Section 3.2), we

leave its investigation for another study. Considering the large uncertainties in the HP ice creep behavior, the most critical work is to determine the viscosity of HP ice at stresses relevant to convection, between 1 and 100 kPa (Kalousová et al., 2018). The viscosity of ice V relative to that of ice VI was investigated by both groups. Sotin and Poirier (1987) noticed that ice V is harder to deform than ice VI and that the viscosity ratio is close to two orders of magnitude at the experimental conditions. On the other hand, Durham et al. (1996) found that ice V and ice VI have similar viscous deformation behavior. This difference in the experimental results has led us to investigate viscosity ratios of up to 2 orders of magnitude.

Tobie et al. (2005) suggest values of tidal dissipation in the HP ice layer on the order of 3×10^{-10} W/kg for a viscosity of 10^{14} Pa.s. In our reference model, we assume viscosity of 10^{15} Pa.s and thus we use smaller values of dissipation power since it decreases with increasing viscosity. Also, Tobie et al. (2005) employed a Maxwell model whereas an Andrade model, that predicts higher dissipation at higher viscosity, may be more appropriate. Consequently, we investigate the effect of tidal dissipation using values of 5×10^{-11} and 1×10^{-10} W/kg.

Appendix D. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epsl.2020.116416>.

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